DressApp Complete Technical User Manual

Comprehensive user manual and technical reference guide for the DressApp personal wardrobe ecosystem, styling engine, circular marketplace, and administration panels.

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1. Overview & Technology Stack

DressApp is an AI-driven personal wardrobe manager, styling advisor, and circular marketplace. It helps users manage clothing items digitally, auto-crop and tag them, receive weather- and calendar-aware outfit recommendations, scan EU Digital Product Passports (DPP), and trade garments.

Core Value Proposition

- **Digital Wardrobe Ingestion**: Snapshot or upload photo processing with automated background removal, clothing categorization, and attribute tag generation.
- **AI Virtual Stylist**: A conversational agent that contextually reviews your wardrobe, Google Calendar events, and local weather forecasts to suggest daily outfits.
- **Circular Marketplace**: Secure peer-to-peer buying, selling, swapping, and renting of clothes to reduce fast fashion waste.
- **Cost-Per-Wear (CPW) Analytics**: Insights into wardrobe capitalization value, utilization rates, and usage optimization.

Technology Architecture

- **Backend Edge**: Python 3.11 with FastAPI, using asynchronous Motor drivers connected to a MongoDB Atlas cluster.
- **Frontend SPA**: React 19 single-page application utilizing `useSyncExternalStore` custom stores (`stylistStore`, `dailySuggestionsStore`, `useOutfitStore`, `useClosetStore`, `useSuitcaseStore`), Tailwind CSS, Shadcn/UI primitives, and `react-i18next` supporting 12 locales.
- **State & Network Optimization**: In-flight request deduplication, 15-minute store caching, and `visibilitychange` tab revalidation yielding zero background GET requests when idle.
- **Local Machine Learning & Sizing**: CPU-local U2-Net (`rembg`) background matting, SegFormer-b2 clothing parsing, Fashion-CLIP embeddings, and ANSUR II physical body measurement regression model (`body_predictor.py`). Optionally routes to self-hosted GPU containers (SegFormer-b3 + BiRefNet) for fast operations.
- **Conversational STT/TTS**: Real-time client-side Web Speech recognition fallback, multimodal server-side Gemini 2.5 Flash modulations, and on-device offline Piper/Sherpa-ONNX engines.
- **External Integration Services**: OpenWeatherMap API for weather fetching, Google Calendar OAuth for daily schedule exports, OpenStreetMap (Nominatim) address autocomplete, and PayPal Subscriptions/Checkout REST APIs.

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2. Prerequisites

Host Environment Requirements

- **Hardware**: Minimum 4 GB RAM VPS (e.g., Hetzner VPS hosting the production `dressapp.co`).
- **Dependencies**: Docker & Docker Compose stack (including backend, frontend, and Caddy TLS termination).
- **Environment Variables**: API keys configuration (`GEMINI_API_KEY`, `DEEPGRAM_API_KEY`, `OPENWEATHER_API_KEY`, `PAYPAL_LIVE_CLIENT_ID/SECRET`, and Google Calendar OAuth tokens).

User App Requirements

- **Web Browser**: Google Chrome or Apple Safari (required for full voice feature compatibility).
- **Permissions**: Grant Camera permission (for clothing snapshots and QR scans) and Microphone permission (for voice conversation).
- **Network**: Active connection for LLM processing, with IndexedDB caching providing offline catalog browsing.

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3. Step-by-Step Instructions

3.1 Ingesting Garments (Adding Items)

INGESTION PARADIGMS: Photography, EU Product Passports, and Digital Commerce Receipts.

1. Navigate to the **Add Item** screen.
2. Select **Take Photo** (launches native mobile camera) or click **Upload Photos** (opens OS file chooser).
3. The client computes the image's SHA-256 and horizontal difference-hash (dHash) in-browser (~100-180ms) to check against your existing closet.
4. If a match is found, the **Duplicate Preflight Dialog** opens showing matching previews. Select **Skip** or **Add anyway**.
5. Once accepted, the server starts an NDJSON stream. A placeholders preview frame displays within 5-7 seconds, allowing you to edit the item details immediately while the backend completes tagging.
6. Verify auto-detected tags (color, fabric, fit, pattern, occasion). If the cutout shape is incorrect, change the **Category** dropdown; this triggers SegFormer to automatically re-crop the garment.
7. Click **Save** to optimistically paint the item in the closet grid immediately (~16ms) while background WebP thumbnail generation concludes.

1. Tap the **Scan QR (DPP)** button on the Add Item page.
2. Grant camera permissions and align the QR code printed on the garment's retail label, or upload a saved QR screenshot.
3. The backend resolves the URL and executes SSRF safety checks (blocking private IP ranges).
4. The system scrapes the JSON-LD schemas to extract brand, materials composition, supply-chain trace, carbon footprint, and care guidelines.
5. Review the extracted data displayed in the green **Verified DPP Data** accordion panel and click **Save**.

1. **Digital Import Interface**: Open the **Digital Import** tab on the Add Item page.

2. **Sub-mode Selection & Processing**: **Paste Text**: Paste raw email bodies, transactional invoices, or copy-pasted order receipts. This text is sent directly to Gemini multimodal vision models for processing. **Upload Image**: Upload invoice/receipt screenshots or photos. For multi-item receipts, users can drag and resize bounding box crop selectors (`[{id, x, y, w, h}]` represented as container percentages) to crop individual items. An off-screen canvas exports a 90% quality JPEG Blob of the selected region to submit to the backend. The backend executes dual-path concurrent processing: multimodal vision/OCR for transaction details, and SegFormer classification for garment characteristics (color, shape, pattern). **Upload PDF**: Submit store PDF invoices (e.g. Zara or ASOS). The backend uses Gemini vision OCR with column-merging logic to parse items, falling back to local `pypdf.PdfReader` text extraction if offline. Bounding selectors allow line-range cropping on the text output before sending it to the parser. **Web Link**: Input order confirmation links. The backend fetches pages via `httpx` using a browser-like User-Agent and a 15-second timeout safeguard, routing them by content-type (HTML page text parsing vs image vision processing).
3. **Data Merging Rule**: Transaction details (price, brand, size, quantity) are extracted from OCR, taking precedence over vision model inference. Visual attributes (clothing category, secondary colors) are classified by SegFormer.
4. **Receipt-Locked Fields & Saving**: All confirmed details are saved in the database with `from_receipt: true` and their field names recorded in `receipt_locked_fields`. Any subsequent visual re-analysis (e.g. when the user updates the item thumbnail) will never overwrite fields protected in this list. Click **Save** to confirm.

3.2 Conversational AI Virtual Stylist

Describe styling dilemmas and receive hands-free spoken outfit advice.

1. Navigate to the **AI Stylist** screen.
2. Click the microphone icon `[Microphone]` in the chat input bar.
3. Speak your request (e.g., "What top matches my beige trousers for a rainy outdoor lunch?").
4. If Web Speech is supported, your voice transcribes live in the input box. If not, the app records a WebM file and uploads it.
5. The backend routes the voice query to the local Gemma4 container (falling back to Gemini 2.5 Flash transcription if offline).
6. The stylist processes your wardrobe history, local weather forecasts, and calendar events to formulate a styling proposal.
7. The stylist speaks the response using preselected voice profiles (`puck`, `aoede`, or `charon`).
8. Tap **Play reply** (or **Replay** in Hebrew mode) on the card to replay the voice audio.

3.3 Profile, Preferences, and Subsystem Dependencies

The Profile page serves as the core control panel for DressApp. Configuration fields directly impact the performance, routing, and behavior of downstream modules.

1. **Photos & Digital Avatar Stage (`AvatarViewer2D` & `DynamicAvatar`) Why does it matter?**: Renders your visual identity across all try-on canvases using a dual-mode stage (segmented real-body

photo cutout vs dynamic 2D Bezier vector SVG mannequin). **Subsystem Dependencies**: Photo cutouts are background-matted via local U2-Net (`rembg`) and downscaled in-browser to a maximum of 1280px at 82% quality to fit within MongoDB's 16MB document ceiling. The stage applies calibrated positional landmarks (`top-[14.5%]` collar-to-neckline, `top-[36.5%]` waistband-to-waistline, `bottom-[2%]` footwear plane) and proportional chest/hip scaling ($scaleX$). Click *Remove Photo* to instantly switch back to the 2D SVG vector mannequin.

2. **Style Profile (Modest rules, Dress code) Why does it matter?**: It establishes personal boundaries for recommended outfits, preventing the AI from generating inappropriate style suggestions. **Subsystem Dependencies**: The selected parameters (e.g., Modest clothing constraints) are fed directly into the styling prompts for Gemini 2.5 Flash, filtering matching wardrobe outputs before they are shown.
3. **Details (Name, Phone, Occupation) Why does it matter?**: It customizes the communication tone and routes notification alerts. **Subsystem Dependencies**: The user's name is dynamically parsed into emails and system-level pushes. The phone number serves as a fallback registry for scheduled alerts. The occupation parameter is fed to the stylist LLM and Trend Scout personalization ranker to customize proposals.
4. **Body Measurements & Sizing (ANSUR II Regression Model & Sizing Predictor) Why does it matter?**: It eliminates sizing guesswork, allowing automatic retail size calculation, external retail size comparison, and accurate virtual layering. **Subsystem Dependencies**: Entering 4 basic parameters (**Height**, **Weight**, **Waist**, **Foot Length**) triggers the scikit-learn ANSUR II regression model (`body_predictor.py`) to auto-predict 6 structural dimensions (*Shoulders*, *Chest*, *Hip*, *Sleeve*, *Inseam*, *Outseam*). **Deterministic Size Translation**: Once continuous measurements are predicted, the backend sizing engine dynamically converts them into retail clothing sizes: **Shirt Size** (XS-XXL based on chest), **Pants Size** (Waist in inches), **Shoe Size** (US Men/Women and EU standards based on foot length and gender), **Dress Size** (US 0-14+ based on chest, waist, and hips), and **Bra Size** (Band + Cup based on chest and estimated underbust). **Full Auto-Population**: These recommended retail sizes are automatically populated in the *Detailed Edit Mode* fields inside the profile dashboard, filling all 15+ sizing parameters down to shoe size without manual entry. **Integrations**: Measurements are queried directly by the **Shopping Assistant** Chrome Extension content scripts to read size tables on partner websites (Zara, Asos) and recommend sizes.
5. **Lifestyle (Status, Sex) Why does it matter?**: It tailors default recommendations and scores content algorithms. **Subsystem Dependencies**: The sex selection directly affects the ranking logic of daily Trend Scout cards. If a news card category mismatches the user's sex, the algorithm applies a -2.0 score penalty, demoting it in the feed.
6. **AI Configuration (SaaS keys, edge mode, credits) Why does it matter?**: It determines billing routing, operational performance, and network offline status. **Subsystem Dependencies**: Routes text/audio generation queries. Standard setups consume DressApp system credits. Standard accounts operate on a prepaid credit bucket system (`CreditBucket` model with free vs paid credit lists). Daily tasks check and consume credits from the oldest free buckets first (expiring in 30 days) before moving to paid credit buckets (which never expire). Standard free users are replenished with 10 free credits daily (daily reset check). Trial plans include: **Pro (Manager) Trial**: 14 days duration with 50 initial free credits. **Business (Professional) Trial**: 30 days duration with 300 initial free credits and active Campaign slots. If system credits are exhausted, the app enters an async pause-and-resume wait state (up to 60s) checking for top-up events. Inputting personal API keys (Google AI Studio, Anthropic, OpenAI) redirects charges to the user's developer billing accounts. Selecting edge local mode routes queries to the offline Gemma container.
7. **Scheduler & Push (Frequency, daily alarm, style focus) Why does it matter?**: It manages automatic daily style pushes. **Subsystem Dependencies**: Activates `APScheduler` cron jobs on the FastAPI backend. Every morning, it triggers push notifications via `pywebpush` using the client's VAPID keys, matching the selected style focus parameters.

8. **Google Calendar (OAuth sync, export rules) Why does it matter?**: It bridges your wardrobe directly with your real-world calendar events. **Subsystem Dependencies**: Authenticates via Google OAuth. The scheduler queries your calendar to identify events, formats outfits, and pushes calendar events straight to your Google Calendar agenda.
9. **Location Services (GPS tracking, weather accuracy) Why does it matter?**: It coordinates weather-appropriate suggestions and local transaction radius filters. **Subsystem Dependencies**: Triggers `navigator.geolocation` reverse-geocoding. Coordinates are sent to OpenWeatherMap API to adjust the stylist recommendations (e.g., rainwear for downpours). It also calculates distances for local Marketplace listings and experts (e.g.,Lisbon radius checks).
10. **Voice & Language (Virtual stylist voice selection) Why does it matter?**: It establishes text dictionary locales and voice modulations. **Subsystem Dependencies**: Controls the active language for translations via `react-i18next`. The voice selection maps BCP-47 voice codes (e.g., `he-IL` or `ar-JO`) to client Web Speech synthesis voices or offline Piper TTS models.
11. **Invite Friends (Share payload API) Why does it matter?**: It provides a viral loop for free closet expansion. **Subsystem Dependencies**: Appends the referrer's MongoDB ID to the URL. New registrations dynamically query this ID and atomically increment the referrer's `closet_capacity_bonus` by +10 slots, modifying the limit guards in `closet.py`. Free tier capacity is capped at 150 items baseline, but can expand up to a maximum limit of 1000 items through referral credits.

3.4 Wardrobe Migration from Competitors

DressApp provides a streamlined way to migrate your wardrobe from popular competitor applications, eliminating the need to re-photograph every item.

Supported Competitor Apps:

- Whering
- Acloset
- Stylebook
- Smartli
- BeautyAI
- Other (custom input)

Detection mechanism: `'ontouchstart' in window` identifies touch-primary devices (phones, tablets).

Entry point gating — Four locations in the SPA check this flag before allowing migration:

Entry Point	File	Desktop Behavior	Mobile Behavior
Auto-trigger (new user)	`AppLayout.jsx`	Renders migration modal	Shows toast.info (8s)
"Import Wardrobe" pill	`Profile.jsx`	Opens migration modal	Shows toast.info (8s)
Shopping Assistant section	`Profile.jsx`	Renders draggable bookmarklet link	Shows info box with desktop guidance
"Migrate" button	`LoginClosetReminderModal.jsx`	Opens migration modal	Shows toast.info (8s)

Mobile guidance message: "Wardrobe import is available on the desktop version of DressApp. Please open your account on a desktop browser to continue."

Rationale: The `getDisplayMedia` API (required for tab-level screen capture) is not supported on mobile browsers. The bookmarklet installation flow (drag to bookmarks bar) is also desktop-only.

The bookmarklet is a self-contained `async` IIFE encoded as a `javascript:` URI that runs on the competitor's page.

Initialization Phase:

1. Remove Chrome extension widgets that conflict with migration
2. Send message to DressApp Chrome extension
3. Inject heartbeat animation CSS
4. Create fixed overlay widget with "Share & Start Agent" button

Screen Capture Phase:

1. Request tab-level screen share via `navigator.mediaDevices.getDisplayMedia()`
2. Primary: `{ video: { displaySurface: 'browser' }, preferCurrentTab: true }`
3. Fallback: `{ video: true, preferCurrentTab: true }`
4. Validates tab-level capture (rejects "window" or "screen" share modes)
5. Hidden `<video>` element receives MediaStream, waits for readyState >= 3

Card Detection (Dual-Strategy DOM Scanner):

Strategy 1 — `<img>` candidates:

- Query all `<img>` elements on the page
- Filter: minimum 40×40px, within viewport bounds, not inside `<header>`, `<nav>`, or `<footer>`
- Walk up DOM to find card-shaped ancestor (width 80–600px, height 80–800px, aspect ratio 0.7–2.5)
- Validate image fills at least 50% of card area
- **Fully-visible gate**: `r.top >= 0 && r.bottom <= innerHeight` — rejects partially clipped cards

Strategy 2 — `<div>` fallback:

- When no `<img>` cards found, scan `<div>`, `<li>`, `<a>`, `<section>` elements for card-shaped containers with CSS `background-image` or nested `<img>` elements

Deduplication: Coordinate-based rejection — two rects within 20px are considered duplicates. *Row grouping:* Cards grouped by Y-center tolerance of 30px for row-aware scrolling.

Crop Engine (`cropCardFromStream()`):

1. Maps CSS viewport coordinates → video pixel coordinates via `scaleX = videoWidth / innerWidth`
2. Clamps source rectangle to video bounds
3. Draws to off-screen `<canvas>`
4. **Quality gate**: Computes color variance — rejects crops with variance < 2
5. **Aspect ratio gate**: Rejects crops where width > 5× height or vice versa
6. Outputs JPEG base64 at quality 0.85

Autoscroller (Row-Aware Scroll Loop):

1. Hide overlay → wait 300ms + double-rAF (layout settle)
2. Detect fully visible cards via `getVisibleGarmentRects()`
3. For each new card: crop from video stream → store
4. Every 15 cards: postMessage(STREAM batch) to parent window
5. Show overlay with card count

6. Calculate scroll amount (row-aware, based on grouped row heights)
7. Scroll, wait 400ms + double-rAF
8. Center the middle row on screen
9. Adaptive lazy-image poll (12s timeout, 600ms intervals)
10. Terminate when: no change + no new rects + noChangeCount >= 3

Completion Phase:

1. Stop MediaStream tracks (release screen capture)
2. Send remainder cards via STREAM message
3. Send DRESSAPP_MIGRATION_COMPLETE with total_cards + app_name
4. Clear harvestedCards array (memory cleanup)
5. Show green completion badge in overlay

A global React component mounted at the App level that survives route navigation.

Message Protocol:

| Message Type | Payload | Behavior | |---|---|---| | **`DRESSAPP_MIGRATION_STREAM`** | **`{ cards: [{crop_base64}] }`** | Appends to **`collectedCards[]`** buffer | | **`DRESSAPP_MIGRATION_COMPLETE`** | **`{ total_cards, app_name }`** | Saves to DB + starts Stylist worker |

Data Flow (Atomic Single-Call Pipeline):

1. Collect cards from STREAM messages → **`collectedCards[]`**
2. On COMPLETE: Filter: **`collectedCards.filter(c => c.crop_base64)`** Clear buffer: **`collectedCards.length = 0`** Single atomic call: **`api.saveMigrationCrops({app_name, cards})`** Navigate: **`navigate('/closet')`**

The core production endpoint that atomically saves crop images to MongoDB and kicks off Stylist analysis in a single request.

Request:

```
class MigrationSaveCropsIn(BaseModel):
    app_name: str = "Competitor App"
    cards: list[dict[str, Any]]  # Each must have crop_base64 (base64 JPEG)
```

Processing:

1. For each card: Extract crop_base64 (raw base64 or data URL with prefix) Decode to raw bytes Validate minimum size (>100 bytes) Validate aspect ratio (width/height between 0.2 and 5.0) Build data URL for segmented_image_url Create ClosetItem (brand=app_name, source="Private", category="Top") Compute source_phash and source_color_sig via image_hash service Insert into MongoDB
2. If items saved AND garment_vision_service configured: Snapshot saved items by ID list (avoids race condition) Spawn **`asyncio.create_task(_run_reanalyze_items)`** Return job_id for status tracking
3. Return **`{items_saved, items_skipped, item_ids, app_name, job_id}`**

Response:

```
{
  "items_saved": 23,
  "items_skipped": 2,
  "item_ids": ["item_a1b2c3d4e5f6", ...],
  "app_name": "Whering",
  "job_id": "reanalyze_a1b2c3d4e5f6"
}
```

A shared async function that processes closet items through Stylist (Gemini) analysis. Runs as an `asyncio.create_task` — completely independent of the client.

Processing Per Item:

1. Resolve best image URL (priority chain): segmented_image_url → cutout_url → clean_image_url → image_variants.webp.large → image_variants.webp.medium → reconstructed_image_url → image_variants.original → original_image_url → image_url
2. Download image bytes via `_read_image_bytes_from_url()`
3. Run Gemini analysis under `_ANALYZE_LOCK` (semaphore=1)
4. Apply `_safe_analysis()` normalization
5. Filter unidentifiable items via `_is_unidentifiable()`
6. Build update_doc with Stylist fields: title, name, caption, category, sub_category, item_type, brand, gender, dress_code, season, tradition, colors, fabric_materials, pattern, state, condition, quality, repair_advice, tags
7. Mirror dominant color/material into legacy scalars
8. Update MongoDB via `repos.update()`

Standalone endpoint for manually re-triggering Stylist analysis on all items matching a brand.

```
class MigrationReanalyzeIn(BaseModel):
    app_name: str = "Competitor App"
```

The modal UI that guides users through the migration flow:

1. **Ask step** (`step === 'ask'`): Welcome dialog with "No, Start Fresh" / "Yes, Import Closet"
2. **App search step** (`step === 'app_search'`): Preset grid (Whering, Acloset, Stylebook, Smartli, BeautyAI) with manual app name input
3. **Web login step** (`step === 'web_login'`): Bookmarklet install instructions, "Import wardrobe" button opens popup via `window.open(targetLoginUrl, '_blank')`

Mobile/Desktop Branching:

- **Desktop**: Renders a draggable `<a>` element with `href="javascript:..."` — user drags to bookmarks bar
- **Mobile**: Renders a hidden `<textarea>` with the bookmarklet code, uses `document.execCommand('copy')` to copy to clipboard

A `useSyncExternalStore`-compatible global store that tracks:

- **Analyze jobs**: In-flight `/analyze` requests (registered by AddItem.jsx)
- **Polish jobs**: Pending background matte/reconstruction tasks

The global poller runs at 3s intervals and checks polish pending items via `api.getItem(id)` with a 5-minute timeout per item.

A draggable glass-morphism pill positioned at `bottom-20` on mobile and `bottom-4` on desktop. Subscribes to `workStore` via `useSyncExternalStore` and renders:

- **Analyzing**: "Analyzing N photos..." with pulsing sparkle icon
- **Done**: Brief "✓ Done" linger (1.2s) before auto-hiding

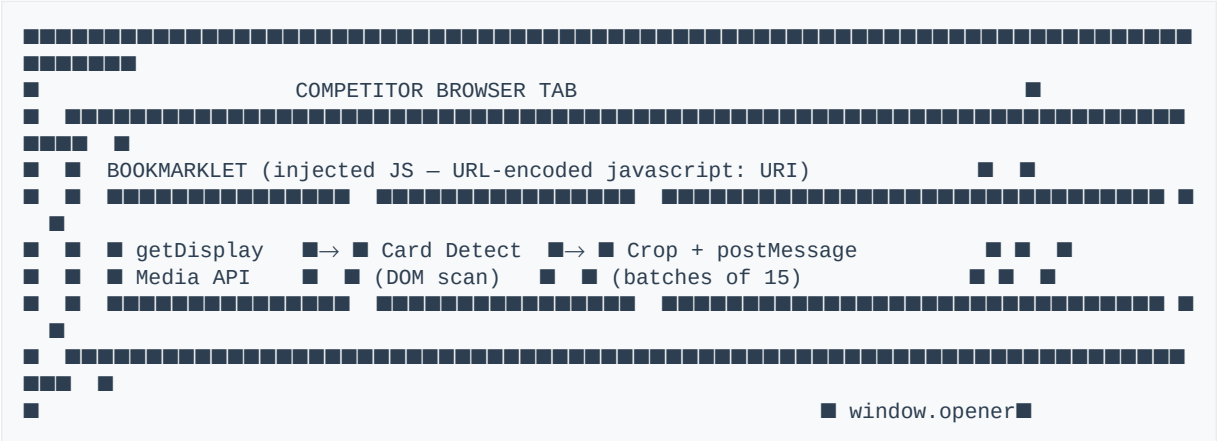
1. **Access Migration:** From your Profile page, click the **"Import Wardrobe"** button (desktop only)
2. **Select Source:** Choose your competitor app from the grid or enter a custom app name
3. **Login to Competitor:** A new browser tab will open to your competitor's login page

4. **Install Bookmarklet:** Drag the "Share & Start Agent" button to your browser's bookmarks bar
5. **Navigate to Wardrobe:** Go to your wardrobe/closet view in the competitor app
6. **Run Migration:** Click the installed bookmarklet button
7. **Automatic Capture:** The agent will: Scroll through your wardrobe Detect and crop garment images Stream captured items back to DressApp in batches of 15
8. **AI Analysis:** Each captured garment is automatically analyzed by our Stylist AI, which identifies: Item type and category Color palette Fabric/material Brand (where visible) Season and occasion Dress code level
9. **Review & Edit:** Once migration completes, review your imported items in the Closet tab

- Uses `getDisplayMedia` API for tab-level screen capture
- Items are processed server-side, so you can close the competitor tab once migration starts
- Duplicate detection prevents importing the same item twice
- Unidentifiable items (obscured images, non-garment items) are automatically filtered out
- Mobile browsers are not supported for migration (use desktop instead)
- Stylist worker runs server-side — closing browser has no effect once save-crops returns

1. **Bookmarklet card detection**: Requires cards to be fully visible in viewport. Non-standard HTML (CSS backgrounds, canvas, shadow DOM) may not be detected
2. `_is_unidentifiable` **filter**: Drops crops where Gemini returns empty analysis, give-up phrases in title/caption, or missing `item_type` + `sub_category`
3. **Gemini API failures**: Individual `analyze()` calls may fail with quota/timeout errors — items are skipped
4. **Browser closure during bookmarklet crawl**: If DressApp tab is closed while bookmarklet is running, crops are lost
5. **Browser closure after save-crops**: Safe — Stylist worker runs server-side

- If the bookmarklet doesn't appear, ensure your bookmarks bar is visible (Ctrl+Shift+B)
- For best results, ensure all wardrobe items are fully visible in the browser window
- If migration stalls, try refreshing the competitor page and restarting the bookmarklet
- Check the DressApp console for diagnostic messages if items aren't appearing
- **Manual re-trigger:** Use the backend endpoint to re-analyze all items from a specific competitor:`curl -X POST -H "Authorization: Bearer <token>" \ -H "Content-Type: application/json" \ -d '{"app_name": "Whering"}' \ https://dressapp.co/api/v1/closet/migration/reanalyze-by-brand`



| Resource | Limit | Rationale | |---|---|---| | `_ANALYZE_LOCK` | 1 concurrent request | Prevents Gemini quota contention | | Scroll retry | 3 attempts | Prevents infinite loops on short pages | | Lazy-image poll | 12s timeout, 600ms intervals | Balances completeness vs. speed | | Bookmarklet batch size | 15 cards per STREAM message | Memory management in postMessage | | Status cleanup | 60 seconds after completion | Prevents memory leak from job metadata |

3.5 Wardrobe Insights Dashboard

Analyze wardrobe capitalization value, tracking garment utilization, and cost-per-wear parameters.

1. Navigate to **Wardrobe Insights**.
2. **Review Metrics**: *Closet Worth*: Dynamic summation of purchase prices. *Closet Utilization*: Percentage of closet garments worn at least once. *Average Cost-per-Wear (CPW)*: Computed as `Price / Wear Count`.
3. **Distribution Graphs**: Toggle tabs to view Recharts visualizations: *Color Palette*: Mapped hex codes distribution. *Materials*: Fabric percentages distribution. *Subcategories*: Mapped subcategories.

4. **Efficiency Leaderboard**: View the top 5 garments showing the lowest Cost-per-Wear scores.

3.6 Outfit Canvas & Planner

Build, layer, and review outfit proposals on an interactive 2D avatar canvas.

1. Open the **Outfit Canvas** planner.
2. **Outerwear Layering (Dual Canvas)**: If your outfit includes outerwear (e.g., a jacket) over a top, the page renders two vertical canvas modules: "With Outerwear" (showing the jacket layered) and "Without Outerwear" (revealing the top underneath).
3. **Interactive 2D Elements**: Tap directly on any clothing piece on the avatar's body. The app routes you straight to that garment's detail screen.
4. **Review Metrics Tab**: Click the details button and choose the **Metrics** tab to view progress bars for compatibility criteria: *Color Harmony* (neutral harmony) *Pattern Compatibility* (pattern clashing prevention) *Body Fitting* (size matching) *Weather Match* (season suitability) *Event Match* (activity appropriateness) *Location Match* (modest rules checks)
5. **Rename/Describe**: Click the Pencil icon to edit outfit names and descriptions.

3.7 Suitcase Assistant

Organize your packing requirements for trips without overpacking.

1. Go to the **Suitcase** page and fill out the Trip Context form (destination, start/end dates, trip category, calendar events).
2. The AI generates a customized packing list and daily outfits based on trip duration and weather forecasts.
3. Review the packing progress. If an important item is missing (e.g., umbrella for rain, swimwear for beach), the system alerts you and suggests matches from the marketplace or local stores.
4. Use the integrated chat box to refine suggestions (e.g., "Change day 2 to casual evening wear"). The assistant edits the suitcase while preserving the rest of the list.
5. Tap **Approve Suitcase** to finalize your plan.

3.8 Scheduler & Push Reminders

Set daily styling alerts to receive outfit recommendations automatically.

1. Open **Profile** and go to **Scheduler & Push**.
2. Toggle notifications, set a daily notification time, weekday frequency, and styling focus theme.
3. Every morning, the background cron task (`APScheduler`) checks weather forecasts and sends a push notification.
4. Tap the notification on your device (or view the web app's Notification Center) to open a proposal dialog displaying 3 styled suggestions.

5. Save a suggestion directly to your **Wardrobe Diary**.

3.9 Marketplace (Resale, Renting, Swapping, Gifting)

Engage in the peer-to-peer circular fashion marketplace.

- **Create a Listing**: Open an item's detail page, select **Edit Intent**, and choose a non-private intent: *For Sale*: Input list price and currency (detects your default currency via regional locale preferences). *Rent*: Set daily rental tariff and borrowing conditions. *Swap*: Mark item open for trade. *Donate*: Publish item for free.
- **State Synchronization**: Listings propagate to the feed automatically. The client uses `useSyncExternalStore` and IndexedDB caching to load search parameters with zero-latency.
- **Try-On Sandbox**: Renters/buyers can test-fit a listing against items in their private closet before checking out.
- **Transactional Checkout**: *Purchasing/Renting*: Complete transaction via integrated PayPal buttons. Captured webhooks notify the seller, flip listing state to sold/rented, and log transactions in the ledger minus the 7% platform fee. *Bartering (Swapping)*: Prospective swappers propose trades. Lister receives confirmation emails to accept or decline.

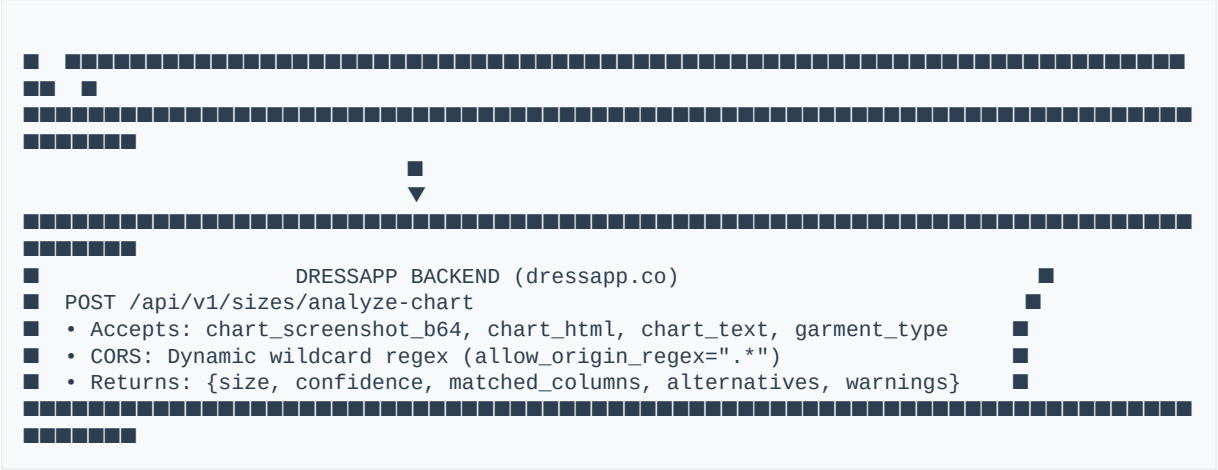
3.10 Shopping Assistant Chrome Extension & Mobile Bookmarklet

A cross-platform client-side utility that resolves sizing uncertainty on third-party e-commerce storefronts by overlaying personalized size recommendations directly on product pages.

The DressApp Extension Ecosystem operates in two modes:

| Mode | Runtime | Screenshot API | DOM Extraction | Storage | |---|---|---|---|---| | **Chrome Extension (MV3)** | Background Service Worker | `chrome.tabs.captureVisibleTab` | Direct DOM access | `chrome.storage.local` | | **Mobile Bookmarklet** | Page sandbox (IIFE) | Not available | IoU-based DOM fallback | `localStorage` with `dressapp_widget_` prefix |

Dual-Mode Routing Architecture:



Multi-Step Heuristic Scoring Algorithm:

1. HTML Table Scanner:

- Iterates over all `<table>` elements in the DOM
- **Base Checks**: Must contain at least 1 number and be longer than 40 characters
- **Keyword Hits**: Adds `+8` points for every matching measurement string (e.g., `bust`, `waist`, `inseam`)
- **Row Count Boost**: Adds `+5` points per row (capped at `+50`)
- *Threshold*: Tables scoring ≥ 30 are selected

2. Image Chart Scanner:

- Evaluates all visible `<img>` elements based on layout and metadata rules
- **Aspect Ratio**: Rejects images with ratio < 0.4 or > 4.0 (typically thin side-banners)
- **Area Limit**: Must be at least 160 × 120 px
- **Semantic Boost**: Adds `+30` points for alt texts or aria labels containing strong keywords (e.g., `"size chart"`)
- **Product Photo Penalty**: Deducts `-14` points if parent classes match typical image gallery keywords (`gallery`, `carousel`, `swiper`)

3. Bounding-Box DOM Fallback (IoU Matcher):

- When running as a bookmarklet (no access to extension screenshot APIs), the manual crop draws a CSS-pixel box
- Executes an **Intersection over Union (IoU)** search over the DOM: $$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$
- The element yielding the highest IoU score is selected as the target container
- If this container points to an `<iframe>` on the same domain, automatically extracts the inner `contentDocument.body` contents

1. Auth Handoff (`RECEIVE_HANDOFF`): When a user clicks "Connect to DressApp" on the main portal, the portal broadcasts a custom event:

```
window.postMessage({ type: 'DRESSAPP_EXT_TOKEN', token: '...', backend: '...' }, '*');
```

The content script intercepts this message and forwards it via `sendToBackground` to be securely saved inside Chrome's local storage (or local storage namespaces prefixed with `dressapp_widget_` if running in bookmarklet mode).

2. CORS Bypass Strategy: The FastAPI backend allows arbitrary origins using a dynamic wildcard regex (`allow_origin_regex=".*"`). This allows bookmarklets running on external store origins like Shein to communicate directly with `https://dressapp.co/api/v1` without triggering preflight blocks.

3. Payload Construction: The endpoint `/sizes/analyze-chart` consumes a unified payload:

```
interface AnalyzeChartIn {
  chart_screenshot_b64?: string; // Crop image from canvas
  chart_html?: string;           // Extracted DOM html fallback
  chart_text?: string;           // Extracted raw innerText fallback
  garment_type: string;          // Extracted type hint
  store: string;                 // Domain name
  page_url: string;              // Full URL
  page_title: string;            // Page header
}
```

- **React-Free Footprint**: The overlay UI (`overlay.js`) compiles into standard vanilla DOM creation calls, isolating variables and preventing CSS and JS namespace leakage from crashing parent page scripts
- **Visual Isolation**: All modal elements styled using custom classes mapped inside `content.css` with high-specificity selectors to prevent host page CSS frameworks (Bootstrap, Tailwind) from overriding the widget
- **Dynamic Internationalization**: Custom wrapper around `i18next` that parses document headers to detect Hebrew and Arabic locales, automatically inserting `dir="rtl"` flags to flip layouts and mirror warning elements

Autodetect Flow:

1. Upon page load, the mutation observer scans the page for size guides
2. If a size-selector container is found, the **Inline Anchor Button** (`* DressApp size`) is mounted next to it
3. A **Floating Action Button (FAB)** (`* DressApp`) is also mounted in the bottom-left corner
4. Clicking either button triggers the automatic detection engine: Scans visible dialogs, description tabs, and parent layouts for HTML `<table>` containing size keywords, or large `<img>` matching chart aspect-ratios If found, captures a crop of the table or encodes the image to base64, querying the backend

Manual Crop Flow:

1. If autodetect fails, the interface shifts into **Crop Mode**
2. Semi-transparent dimming overlay covers the webpage with header: `Drag a box around the size chart`
3. **Drawing Gesture**: Left-click and hold, drag diagonally to outline the chart, release to lock
4. **Refining Layout**: Hovering over crop boundaries exposes 8 resize handles (edges and corners)
5. **Applying**: Click **Apply** (or press `Enter`) to confirm the crop
6. **Cancel**: Click **Cancel** (or press `Escape`) to exit crop mode

| State | Content | User Action | |---|---|---| | **Recommendation Card** | Recommended size, confidence index, reasoning text, matched columns, response latency, alternative size fits | Dismiss or retry | | **Warning Card** | Suspected typos in profile (e.g., `"Your shoulders (75 cm) look higher than expected"`) | Edit profile | | **Authentication Card** | Session expired/disconnected, link to `dressapp.co/extension/connect` | Login | | **Missing Measurements Card** | No circumferences or sizes registered in profile | Edit profile | | **No Matches Found** | Gemini reads chart but finds no row matching measurements | Retry with tighter crop | | **Orphaned Scripts** | Extension updated in background, invalidating content script port | Reload tab |

- **No Matches found (`No clear match`)**: Surfaces when Gemini reads the chart but finds no row matching the user's measurements. A **Retry** button allows re-cropping more tightly.
- **Orphaned Scripts (`DressApp was just updated`)**: Occurs when the Chrome extension updates in the background, invalidating the content script's port. Widget detects stale context and prompts reload.

- **Permission Requests**: If a screenshot fails because the extension lacks permissions, triggers browser permission dialog for optional `<all_urls>` permission.

| Resource | Limit | Rationale | |---|---|---| | `_ANALYZE_LOCK` | 1 concurrent request | Prevents Gemini quota contention | | Scroll retry | 3 attempts | Prevents infinite loops on short pages | | Lazy-image poll | 12s timeout, 600ms intervals | Balances completeness vs. speed | | Bookmarklet batch size | 15 cards per STREAM message | Memory management in postMessage | | Status cleanup | 60 seconds after completion | Prevents memory leak from job metadata |

3.11 Admin Panel Dashboard

System liveness validation, financial bookkeeping, and user account management.

1. Navigate to `/admin` (available to administrator roles).
2. **Overview**: Audit Gross Volumes and Platform Fee income summaries. Inspect the **Provider Activity Table** to view downstream liveness statistics (Gemini API, weather service latency, and error rates).
3. **Providers**: Click **Verify Key** to send a direct ping to the Gemini API. Toggle the **Eyes Vision Override** switch to route image analysis between the default Gemini endpoint and a local Gemma container.
4. **Users**: View active credits, roles, and lifetime payments. Use direct actions to Promote or Demote users.
5. **Listings**: View listing states and toggle active flags to suspend fraudulent items.

3.12 Subscriptions, Billing, and Localized Payments (Atzmai Gateway)

Integration with PayPal and Atzmai APIs for subscription management, credit purchases, and automated bookkeeping.

1. **Atzmai Gateway Integration**: Handles local Israeli payments in ILS/USD supporting: **Regular Credit Cards**: Standard online transactions. **Bit Payment Integration**: Direct mobile checkout links. **Recurring Payments**: Standing orders (direct debit equivalents) for monthly/annual subscriptions.
2. **Purchasing Credit Packs**: Credit purchases are initiated by posting a request with a desired plan selection to the checkout router. Price points include: **10 credits pack**: $1.99 / 10.00 ILS **25 credits pack**: $3.99 / 25.00 ILS **50 credits pack**: $7.99 / 50.00 ILS **Custom top-up amount**: User-specified amount in cents (minimum 5.00 ILS threshold).
3. **Webhook Processing & Verification**: Upon payment authorization, the gateway sends a callback containing the transaction details to `POST /api/v1/atzmai/webhook`. The system verifies the order payload: Checks matching database transaction documents (`db.atzmai_topups`). Confirms success status (`captured`). Allocates the purchased quantity to the user's paid credit buckets.
4. **Invoicing & Receipts**: On successful capture, the backend contacts the gateway billing API to retrieve PDF attachments of localized invoices and receipts. These files are dispatched to the user's email address via automated confirmation messages.

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4. Expected Results

- **Ingestion**: Items immediately populate the closet grid (~16ms). Background cutouts render clean, transparent PNG results.
- **DPP Verified Badge**: Scanning valid passports displays the green information card with sustainability details.
- **Avatar Outerwear**: Outerwear displays correctly layered over tops on the 2D avatar canvas without clipping headwear/shoes.
- **Voice Response**: Virtual Stylist text outputs play spoken audio automatically with a visible waveform indicator.
- **Subscriptions**: Activating Pro immediately removes the 150-item limit warning.

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5. Troubleshooting

HTTP 402 Payment Required

- **Problem**: Ingestion blocked. You have reached the maximum baseline limit of 150 closet items.
- **Solution**: Go to Profile -> Subscription and upgrade to Pro, or share your invite link to get +10 slots per registration.

SSRF Blocked / DNS Error on DPP

- **Problem**: Scanned QR passport URL fails to parse.
- **Solution**: The parser blocks private IP addresses (e.g. `127.0.0.1`, `192.168.x.x`) to protect internal servers. Ensure QR codes point to public domains.

Camera / Microphone Permission Denied

- **Problem**: Capture/scan viewport displays an 'X' error screen, or voice typing fails.
- **Solution**: Open browser permissions, enable Camera and Microphone access for the domain, and reload.

Stylist Chat Failure / Rate Limits

- **Problem**: Chat shows errors or freezes.
- **Solution**: The server catches Gemini `429` rate limits and falls back to a rule-based closet selection algorithm. Verify your internet connection.

Out of Memory (OOM) VPS Spikes

- **Problem**: CPU/RAM peaks during upload processes.
- **Solution**: Ingestion utilizes sequential queue locks for batches >5 items. Ensure the server has at least 4 GB RAM.

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6. Limitations

- **Browser Web Speech APIs**: Native Speech-to-Text translation is restricted to Chrome and Safari; other browsers fall back to standard text input.
- **Offline Client Modulations**: Mobile offline Piper ONNX speech synthesis uses fewer voice profiles than the server-side Gemini audio modal.
- **Image Size Constraints**: Avatar and profile uploads are compressed locally in the browser to 82% quality to fit within the 16MB MongoDB document boundary.
- **Receipt Parsing Scope**: Highly blurred, distorted, or handwritten receipts may fail data extraction.